

- Description

This module manages a smart boiler system, ensuring that:

1. **The water temperature inside the boiler is continuously monitored to maintain optimal heating efficiency.**
2. **If the temperature drops below the minimum threshold, the heating system is activated.**
3. **If the temperature rises above the maximum threshold, the heating system is deactivated to prevent overheating.**
4. **A log of all temperature readings, heating system activations, and shutdowns is maintained for energy optimization.**

This system helps improve heating efficiency, prevent overheating, and optimize energy consumption in water heating applications.

- Specification

module SmartBoilerTemperatureControl

var currentTemperature, maxTemperature, minTemperature: real

var heatingActive: boolean

process updateTemperature(changeAmount: real)

functional scenario form:

currentTemperature >= minTemperature && Water temperature maintained.

||

currentTemperature < minTemperature && heatingActive = true, Heating system activated.

||

currentTemperature > maxTemperature && heatingActive = false, Heating system deactivated.

end process

end module

- Java Code

```
package code;
class SmartBoiler {
    private double maxTemperature;
    private double minTemperature;
    private double currentTemperature;
    private boolean heatingActive;
    public SmartBoiler(double maxTemperature, double minTemperature, double initialTemperature) {
        this.maxTemperature = maxTemperature;
        this.minTemperature = minTemperature;
        this.currentTemperature = initialTemperature;
        this.heatingActive = false;
    }
}
```

```

        public void updateTemperature(double changeAmount) {
            currentTemperature += changeAmount;
            manageHeatingSystem();
            System.out.printf("Current Water Temperature: %.2f°C\n", currentTemperature);
            if (heatingActive) {
                System.out.println("Heating System Activated.");
            } else {
                System.out.println("Heating System Deactivated.");
            }
        }
        private void manageHeatingSystem() {
            if (currentTemperature < minTemperature) {
                heatingActive = true;
            } else if (currentTemperature > maxTemperature) {
                heatingActive = false;
            }
        }
    }
}

public class c68_SmartBoilerSystem {
    public static void main(String[] args) {
        SmartBoiler boiler = new SmartBoiler(80.0, 40.0, 60.0);
        // testcase-VT:
        //     boiler.updateTemperature(-25.0);
        //     boiler.updateTemperature(20.0);
        //     boiler.updateTemperature(30.0);
        //     boiler.updateTemperature(-10.0);

        // testcase-FT:
        //     boiler.updateTemperature(-1.360606960895792E308);
        //     boiler.updateTemperature(0.0);
        //     boiler.updateTemperature(4.9E-324);
        //     boiler.updateTemperature(-3.934312560431096E307);

        // testcase-Z3:
        //     boiler.updateTemperature(-25.0);
        //     boiler.updateTemperature(20.0);
        //     boiler.updateTemperature(30.0);
        //     boiler.updateTemperature(-10.0);

        // testcase-UVT:
        boiler.updateTemperature(-25.0);
        boiler.updateTemperature(20.0);
        boiler.updateTemperature(30.0);
    }
}

```

```

        boiler.updateTemperature(-10.0);
    }
}

```

- Coverage report

1. Test case number: 4
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c68_SmartBoilerSystem.java	100.0 %	6	0	6
> c68_SmartBoilerSystem		0	0	0
> SmartBoiler	100.0 %	6	0	6

3. Vibration testing instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,769	20,855
▼ src	0.4 %	86	20,769	20,855
▼ code	0.4 %	86	20,769	20,855
▼ c68_SmartBoilerSystem.java	96.6 %	86	3	89
> c68_SmartBoilerSystem	87.0 %	20	3	23
> SmartBoiler	100.0 %	66	0	66

4. Fuzz testing branch coverage: 33.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.2 %	2	840	842
▼ src	0.2 %	2	840	842
▼ code	0.2 %	2	840	842
▼ c68_SmartBoilerSystem.java	33.3 %	2	4	6
> c68_SmartBoilerSystem		0	0	0
> SmartBoiler	33.3 %	2	4	6

5. Fuzz testing instruction coverage: 83.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	74	20,781	20,855
▼ src	0.4 %	74	20,781	20,855
▼ code	0.4 %	74	20,781	20,855
▼ c68_SmartBoilerSystem.java	83.1 %	74	15	89
> c68_SmartBoilerSystem	87.0 %	20	3	23
> SmartBoiler	81.8 %	54	12	66

6. Z3 branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c68_SmartBoilerSystem.java	100.0 %	6	0	6
> c68_SmartBoilerSystem		0	0	0
> SmartBoiler	100.0 %	6	0	6

7. Z3 instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,769	20,855
▼ src	0.4 %	86	20,769	20,855
▼ code	0.4 %	86	20,769	20,855
▼ c68_SmartBoilerSystem.java	96.6 %	86	3	89
> c68_SmartBoilerSystem	87.0 %	20	3	23
> SmartBoiler	100.0 %	66	0	66

8. Unimproved V-method branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	836	842
▼ src	0.7 %	6	836	842
▼ code	0.7 %	6	836	842
▼ c68_SmartBoilerSystem.java	100.0 %	6	0	6
> c68_SmartBoilerSystem		0	0	0
> SmartBoiler	100.0 %	6	0	6

9. Unimproved V-method instruction coverage: 96.6%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.4 %	86	20,769	20,855
▼ src	0.4 %	86	20,769	20,855
▼ code	0.4 %	86	20,769	20,855
▼ c68_SmartBoilerSystem.java	96.6 %	86	3	89
> c68_SmartBoilerSystem	87.0 %	20	3	23
> SmartBoiler	100.0 %	66	0	66